Scoring, substitution matrices and score aggregation for TCR antigen-specificity search

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vdjmatch technical appendix

Abstract

`vdjmatch` annotates T-cell receptor (TCR) clonotypes by antigen specificity through fuzzy CDR3 search against a labelled database (VDJdb), built on the `seqtree` search core. This note documents the *scoring* layer: how CDR3 similarity is scored, the data-driven TCR substitution matrix (VDJAM) and why it is region-structured, and how per-hit scores are aggregated into a control-calibrated significance. CDR3 loops resist the assumptions behind BLOSUM/PAM — they cannot be multiply aligned across clonotypes of different length, and germline-encoded flanks (`CASS`...) are so over-conserved that a naive matrix conflates germline bias with selection. We give a position-debiased log-odds estimator that isolates the substitution signal, show empirically that this signal lives in the non-template (NDN) core while the V/J flanks are germline-determined, and demonstrate the consequences on worked examples (GIL, NLV) using the alignments and CIGAR strings produced by `seqtree`. The E-value theory used for significance is derived in the companion `seqtree` appendix; here we summarize it and give the paired-chain extension.

1. The search-and-score framework

Given a query CDR3 q and a labelled reference set D (VDJdb), `vdjmatch` retrieves the neighbours of q within a fixed scope $\theta = (s, i, d, t)$ (max substitutions, insertions, deletions, total edits) using the `seqtree` engine, which defines a ball $B_\theta(q) = \{x : \mathrm{edit}(q, x) \leq \theta\}$ and returns, per hit, the edit decomposition and a matrix-weighted alignment. The matrix enters as a *ranking* score inside the ball; the ball itself (an edit budget) bounds the candidate set. Each hit carries its antigen labels, an extended CIGAR (=/`X`/`I`/`D`), and an alignment score. Significance comes from comparing the neighbour count to a matched background control (§ 3).

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2. Substitution scoring and the VDJAM matrix

How BLOSUM is built, and why it does not transfer.. BLOSUM is a log-odds matrix estimated from the BLOCKS database of *ungapped multiple-alignment blocks* of conserved protein regions. Its recipe is: (i) take many trusted blocks — columns of homologous, gaplessly aligned residues; (ii) cluster sequences within a block above a percent-identity threshold and down-weight within-cluster pairs so abundant near-duplicates do not dominate; (iii) count, over all blocks, how often residues a and b co-occur in the *same alignment column*, q_{ab}; (iv) score $s(a,b) = \frac{1}{\lambda}\log(q_{ab}/p_a p_b)$ against the *global* background frequencies p_a. The signal is "which residues substitute for one another within conserved, column-homologous positions," and the background is position-independent.

CDR3 loops break every step. (i) There is no trusted multiple alignment: same-specificity CDR3s differ in length and have no column homology, so "alignment columns" do not exist. (ii) The germline-encoded ends are over-conserved — essentially every human TRB CDR3 begins `CASS` and ends in a short J motif — so a *global* background p_a is wrong: an alanine in the N-terminal `CASS` is germline-fixed and carries no specificity signal, while an alanine in the centre does, yet BLOSUM would score them identically. A single global matrix estimated over all positions thus conflates germline conservation with antigen-driven selection.

What VDJAM changes (point by point).. VDJAM keeps the log-odds idea but replaces the two broken ingredients. (a) *Observation unit*: instead of columns of a multiple alignment, the unit is a *same-antigen single-substitution pair* — two CDR3s annotated to the same epitope, equal length, differing at exactly one position; this is the only "aligned residue pair" obtainable without a multiple alignment, and (like a BLOCKS column pair) it records two residues that are interchangeable without changing function (here, specificity). (b) *Background*: instead of a global p_a, a *position-specific* background $\mathrm{bg}(\cdot \mid L,p)$ (the residue frequency at that length and position) is used, so over-conserved columns are normalized away analytically rather than masked by hand. (c) *Redundancy*: BLOSUM's percent-identity clustering is replaced — partly by the position-specific background (germline-shared residues are "expected" and contribute little), and prospectively by the control-calibrated significance layer (§3) that already discounts convergent public clones; per-clonotype clustering can be added exactly as in BLOSUM if needed. (d) *Scope*: BLOSUM is one global matrix; VDJAM is estimated per chain and per region, because (next paragraph) only the NDN core carries a transferable substitution signal.

Observation unit and estimator.. We replace "alignment columns" with *same-antigen single-substitution pairs*: ordered pairs of CDR3s annotated to the same epitope, of equal length, differing at exactly one position — the only clean substitution signal extractable without a multiple alignment. For a substitution $a \leftrightarrow b$ observed at position p in a CDR3 of length L on chain c, let $\mathrm{bg}(\cdot \mid L,p)$ be the residue frequency at position p among all length-L chain-c CDR3s. With $f(a,b)$ the observed count of $\{a,b\}$ events and

$$e(a,b) = \sum_{\text{event slots }(L,p)} n_{L,p}\left[\mathrm{bg}(a \mid L,p)\,\mathrm{bg}(b \mid L,p) + \mathrm{bg}(b \mid L,p)\,\mathrm{bg}(a \mid L,p)\right], \qquad \mathrm{VDJAM}(a,b) = \log_2 \frac{f(a,b)+\alpha}{e(a,b)+\alpha}, \tag{1}$$

where $n_{L,p}$ is the number of observed events at (L, p) and α a Laplace pseudocount. The key point is that the background is *position-specific*: at an over-conserved column the residue frequency is peaked, so substitutions there are both rare (small f) and "expected" (matched e), and contribute ≈ 0 to the log-odds — the germline bias is removed analytically rather than by hand-masking. The matrix is consumed by `seqtree` via the squared-distance penalty $\mathrm{pen}(a, b) = s_{aa} + s_{bb} - 2s_{ab}$ (`SubstitutionMatrix.from_similarity`); a dominant uniform diagonal makes higher learned similarity rank a substitution better.

Region structure: NDN is learned, V/J are germline-determined.. A CDR3 is V-germline prefix · NDN insert · J-germline suffix. Estimating (1) separately by region shows (Fig. 1) that the substitution signal — the agreement with general amino-acid chemistry, measured as correlation with BLOSUM62 — is concentrated in the NDN core (TRB $r = 0.42$, TRA 0.35), whereas the V flank carries almost none (TRB 0.08, TRA 0.07): V-flank "substitutions" are dominated by germline allele differences, not free chemistry. This motivates a two-part model: **VDJAM is learned for the NDN core**, while the **V and J flanks are auto-computed** from the germline V/J sequences and their trimming. Concretely, a CDR3 position derived from V- (or J-) germline is near-invariant when it survives trimming and reverts to the random-insert background when trimmed; so the V/J contribution is a per-position conservation profile $w(p) = \Pr[\text{position } p \text{ is germline-retained}]$, determined by the gene's trimming-length distribution, rather than a substitution matrix learned from data. Combined, the NDN matrix scores the core and $w(p)$ down-weights substitutions at germline-retained flanks (realized through a per-position weighting of the base matrix). The germline-retention profile $w(p)$ is computed from the OLGA recombination model (per-gene V/J deletion distributions and germline CDR3-portion segment lengths, via mirpy's `mir.basic.trimming`) and bundled with `vdjmatch`; Fig. 2 shows it drops monotonically from each anchor (the `CASS` V-stem and the J motif are germline; the core is insert). Region-aware scoring weights each CDR3 column by $1 - \Pr[\text{germline}]$ and applies the NDN matrix — implemented in `vdjmatch.match.regions` and evaluated below.

3. Score aggregation and significance

Per-hit to per-query.. A region-aware similarity score sums the NDN matrix score over the core and the (down-weighted) flank contributions; this is available as an optional ranking score and generalizes the legacy cloglog generalized-linear aggregation over V/CDR1/CDR2/J/CDR3.

Control-calibrated E-value (primary).. Immune repertoires are biologically redundant, so an i.i.d. null over-calls. `vdjmatch` instead counts a query's VDJdb neighbours within the ball and compares to the count expected from a matched background control: with $N = |D|$, M the control size and n_C the control neighbour count, $\mathbb{E} = (N/M)\, n_C$ is the expected number of VDJdb neighbours by chance and $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\mathbb{E}) \geq n_D)$ is the enrichment significance — significant only when a clonotype has *more* VDJdb neighbours than the generative background predicts. The Poisson/Chen–Stein theory, the punctured (self-excluding) null and the choice of M are derived in the companion `seqtree` appendix.

Paired $\alpha\beta$ E-value.. The null factorization is not an approximation of convenience: α and β chains pair *almost without constraint* in the repertoire at large, as shown by integrating structural

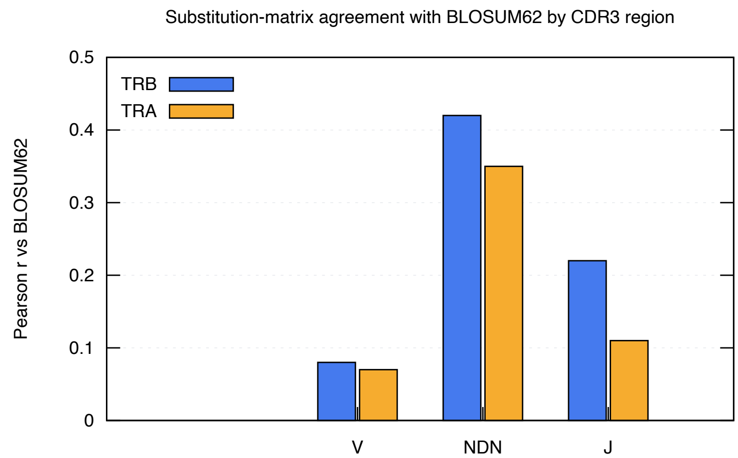

Figure 1: Per-region agreement of the learned CDR3 substitution matrix with BLOSUM62 (Pearson r of off-diagonal scores), human VDJdb. The free-substitution signal lives in the NDN core; the V flank is germline-determined.

and paired-sequencing data [4], so under the background (generative) law the joint ball mass genuinely factorizes, $\pi_0^{\alpha\beta} \approx \pi_0^{\alpha}\pi_0^{\beta}$. Crucially, the *same* analysis finds that *antigen-driven selection distorts* this uniform pairing landscape — i.e. conditioned on an epitope the two chains are *not* independent. This is exactly the structure the joint E-value exploits: the null factorizes (background pairing is unconstrained), so among the N paired VDJdb complexes the expected number of chance joint matches is $\mathbb{E}_{\alpha\beta} = N\,\pi_0^{\alpha}\pi_0^{\beta}$ with p the Poisson tail on the count of complexes matched in *both* chains; an excess over this factorized null is precisely the epitope-conditional dependence (co-selection of α and β) that signals a true specificity. Because the joint null is far smaller than either chain's, true pairs are dramatically more significant: on the paired TCRvdb benchmark, $296/300$ pairs match a complex and $295/296$ (99.7%) recover the correct epitope, with joint p down to 3.6×10^{-9}. (When the two chains' hits are treated as independent tests, the equivalent statistic is Fisher's combination of the per-chain enrichments; the appendix `seqtree` co-occupancy term quantifies the residual dependence.)

4. Results: does VDJAM (and region weighting) help?

A matrix derived from same-antigen pairs must be tested on epitopes it never saw, or the evaluation is circular. For each held-out epitope E^* we re-derive the NDN matrix from *all other* epitopes, fix the candidate set with a single unit-cost search on E^* (exact self excluded), and re-score each (query, hit) pair four ways — unit, BLOSUM62, NDN-VDJAM, and region-weighted NDN-VDJAM — ranking by score and reporting micro PR-AUC for "hit shares E^*" (Fig. 3, eight largest TRB epitopes). Two honest findings. (i) *Region weighting is a real, consistent, but small improvement over the flat matrix* ($+0.004$ at scope 2, $+0.007$ at scope 3; never worse on any epitope, and the gain grows with scope) — modest because CDR3 variation is *already* NDN-concentrated, so down-weighting the rarely-varying germline flanks moves few hits; it should matter more for cross-V-gene comparisons, where flank (germline-allele) differences are common. (ii) *BLOSUM62 remains the strongest substitution*

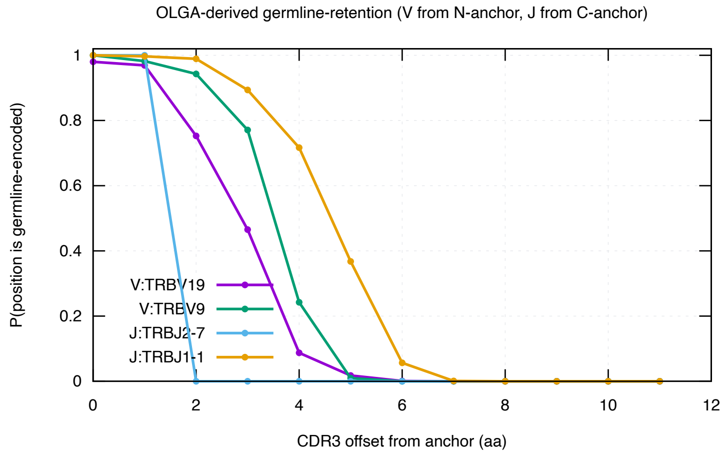

Figure 2: OLGA-derived germline-retention $w(p)$ (via `mirpy`): probability that each CDR3 offset from the V-Cys (or J) anchor is germline-encoded. It falls monotonically into the insert core, defining the region weight $1 - w(p)$ used in scoring.

matrix at this narrow-scope retrieval task (mean PR-AUC 0.32 vs 0.28 region-VDJAM vs 0.29 unit): the data-derived NDN matrix captures general substitution tolerance that BLOSUM already encodes well, and with limited same-antigen pairs it is noisier. At scope 2 the search ball supplies most of the specific-vs-nonspecific discrimination and any matrix only re-ranks within it; the substitution model is therefore a second-order lever here, and the first-order signal is the control-calibrated neighbour count (§ 3).

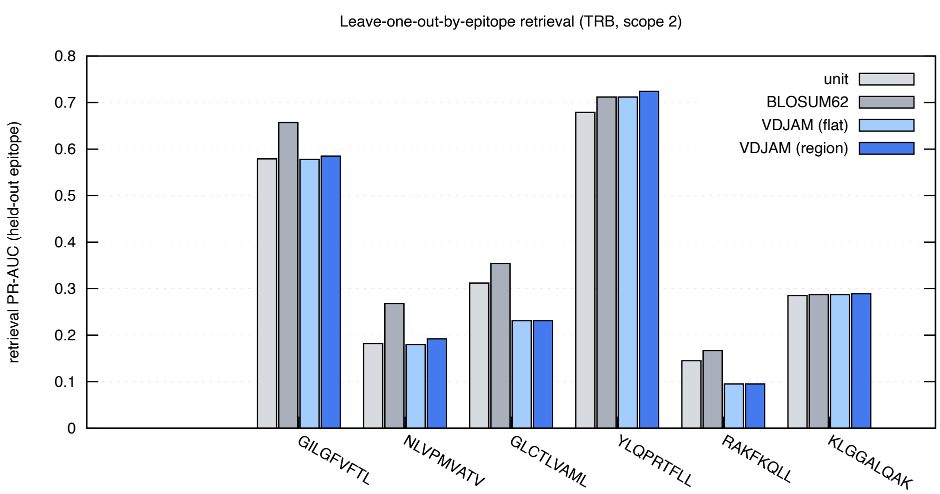

Figure 3: Leave-one-out-by-epitope retrieval PR-AUC (TRB, scope 2): the NDN VDJAM matrix versus BLOSUM62 and unit cost on epitopes excluded from matrix estimation.

5. Edit distance, position, and the matrix comparison

Four analyses (`bench/scoring_analysis.py`, `bench/loo_vdjam.py`) settle what scoring actually buys on VDJdb. All figures and numbers below are the *human TRB* example case (the largest, best-populated slice of VDJdb); the same pipeline applies unchanged to human/mouse and to TRA/TRB — point `$VDJDB_SAMPLE`/`$VDJDB_SPECIES` at another export to regenerate.

Hamming distance 1 is the signal:noise optimum.. The fraction of CDR3 neighbours that share an epitope — the precision of a similarity call — falls steeply with edit distance: 0.53, 0.27, 0.19, 0.15, 0.13 at distance $1 \dots 5$ (Fig. 5). A distance-1 neighbour is twice as likely to be a true co-specific as a distance-2 one and four times a distance-5 one. This reproduces the original VDJdb observation [6] that single-mismatch neighbours are the sweet spot, and it frames everything below: the search ball, not the substitution matrix, is the dominant lever; the matrix only re-ranks within a ball whose purity is already set by its radius.

Central substitutions carry the specificity signal.. Conditioning on *where* a single substitution falls (Fig. 6), the probability that the two CDR3s still share an epitope dips to ≈ 0.41 in the NDN core but rises to ≈ 0.70 toward the J border: a central mismatch most often *changes* specificity (it is contact-proximal and informative), while a border mismatch is usually germline noise (the conserved V/J stems). This is the empirical basis for weighting the centre more — the same direction as the germline-retention weight $1 - w(p)$ (§ 2) — and, encoded directly as a positional weight on BLOSUM62 (the PSSM of § 5), it is the one change that consistently improves retrieval over flat BLOSUM62 (Table 1). It also matches the contacting-region picture of Shcherbinin et al. [5].

The NDN core is glycine-rich.. Central CDR3 positions reach $\sim$ 30–36% glycine in both chains (Fig. 7) — the N-region/D-gene insertion signature — a strongly non-BLOSUM composition. Notably TRA is comparably central-G-rich to TRB, so this is an insertion-diversity feature, not a TRB-D-gene-exclusive one; either way the NDN alphabet is distinctive enough that a general matrix is a poor fit there in principle.

No *standard* matrix beats BLOSUM62.. Re-scoring the held-out-epitope candidates (leave-one-out, § above) with each substitution matrix gives the same verdict at every scope (Table 1, Fig. 4): BLOSUM62 and PAM250 are the best and essentially tied, the structural matrix sits between them and plain Hamming, Hamming (unit) is a strong baseline, and the data-derived NDN-VDJAM — flat or region-weighted — trails them all (region beats flat by a small, consistent, scope-growing margin). So among off-the-shelf and data-derived *amino-acid* matrices, the honest answer to "can we beat BLOSUM62" is no: the substitution matrix is a second-order lever once the ball radius is fixed, and the first-order statistic remains the control-calibrated neighbour count at distance 1 (§ 3).

But position-weighting BLOSUM62 does (the significance PSSM).. Experiment 2 says *where* a substitution falls matters more than *which* substitution it is. We encode that directly as a positional factor on top of BLOSUM62 rather than as a new alphabet matrix. From the position-significance curve (Fig. 6) we take a per-relative-position weight $\omega(p) = (1 - \Pr(\text{same} \mid \text{sub at } p))$, normalised to mean 1 (centre ≈ 1.4, J border ≈ 0.7), and interpolate it to any CDR3 length. `seqtree` consumes this natively: `regions.significance_pssm()` builds a fixed-width `PositionalMatrix.from_weights(BLOSUM62,`

$\lfloor 100\,\omega(p)\rceil$), so the engine scores $\mathrm{pen}(p,a,b) = \omega(p) \cdot \mathrm{BLOSUM62}(a,b)$ — a central mismatch costs $\sim 1.4\times$ its border-position cost. This is the only scoring change that beats flat BLOSUM62, and it does so *at every held-out epitope*: $0.333 \to 0.356$ at distance ≤ 2 and $0.218 \to 0.236$ at distance ≤ 4, winning 8/8 epitopes at both radii (Table 1). The gain is modest in absolute PR-AUC but uniformly signed, exactly because the matrix re-ranks within a ball whose purity is already set by the radius. The remaining lever, the *V-gene* dimension, is taken up in § 6.

A purpose-built TCR matrix (tcrBLOSUM) does not transfer.. Postovskaya et al. [3] report that tcrBLOSUM — a BLOSUM-style matrix counted from same-epitope, equal-length CDR3 pairs — improves epitope-specific TCR clustering over BLOSUM62. Two features make that gain suspect: the counts apply *no* redundancy clustering (so convergent public clonotypes dominate the statistics), and the matrix is validated on the same same-epitope relation it is fit to. Run through our leave-one-out-by-epitope retrieval (their published $\mathrm{tcrBLOSUM_b}$, `bench/tcrblosum_refute.py`), it does *not* beat BLOSUM62: it is slightly worse at both radii (0.303 vs 0.330 at distance ≤ 2, winning 0/8 epitopes; 0.197 vs 0.211 at distance ≤ 4), while their bundled BLOSUM62 reproduces ours exactly (a sanity check on the loader). Reweighting *either* matrix by the central-significance PSSM helps, and it helps BLOSUM62 more than tcrBLOSUM (0.354 vs 0.336). The lesson is consistent: a re-counted amino-acid alphabet does not transfer across epitopes; position does.

mean retrieval PR-AUC	unit	BLOSUM62	PAM250	structural	VDJAM	VDJAM-region	BLOSUM+possig
edit distance ≤ 2	0.306	0.333	0.329	0.314	0.284	0.287	**0.356**
edit distance ≤ 4	0.203	0.218	0.218	0.192	0.177	0.181	**0.236**

Table 1: Substitution matrices compared by leave-one-out-by-epitope VDJdb-vs-VDJdb retrieval (TRB, 8 largest epitopes, micro PR-AUC). Among standard/data-derived amino-acid matrices BLOSUM62 $\approx$ PAM250 lead and VDJAM trails; the only scoring that beats BLOSUM62 is BLOSUM62 reweighted by the positional-significance PSSM (centre > V/J borders), which wins all 8/8 held-out epitopes at both radii.

6. The V-gene dimension

The remaining lever is the V gene. Two questions, both tested by leave-one-out retrieval on human TRB (`bench/vgene_strat.py`, edit distance ≤ 4): is a V match itself a specificity signal, and can it be *softened* to germline-similar V genes?

A V match is a strong, near-binary specificity prior.. Splitting every scored neighbour pair by whether query and hit use the same V family, same-V neighbours share the held-out epitope 53% of the time versus 12% for cross-V — a $\sim$4.3× prior, far larger than any substitution-matrix effect (Table 2). This is why the legacy tool used V as a hard filter, and it argues for carrying V as a strong prior (a soft $S(V)$ term or a per-V null) in a joint V+CDR3 score rather than discarding it.

Position weighting helps both strata equally.. The central-significance PSSM and the germline-retention weighting each add $\approx +0.02$ PR-AUC over flat BLOSUM62 in *both* strata (Table 2). We had expected the gain to concentrate in cross-V pairs, where germline V-flank residues differ and

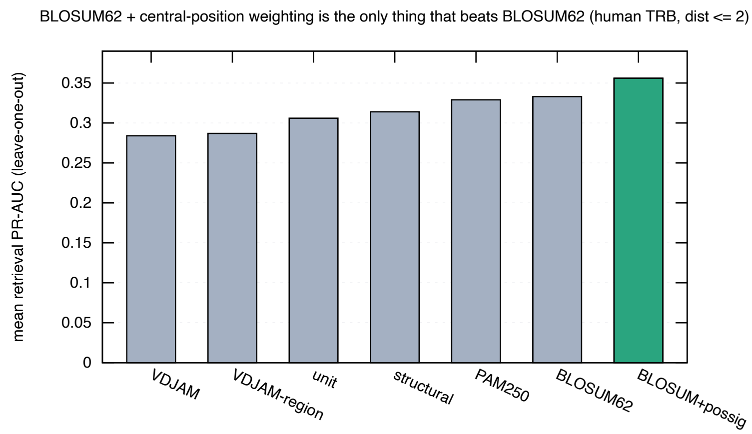

Figure 4: Mean leave-one-out retrieval PR-AUC by scoring (TRB, distance ≤ 2). No standard amino-acid matrix beats BLOSUM62; reweighting BLOSUM62 by the central-position significance factor (green) does.

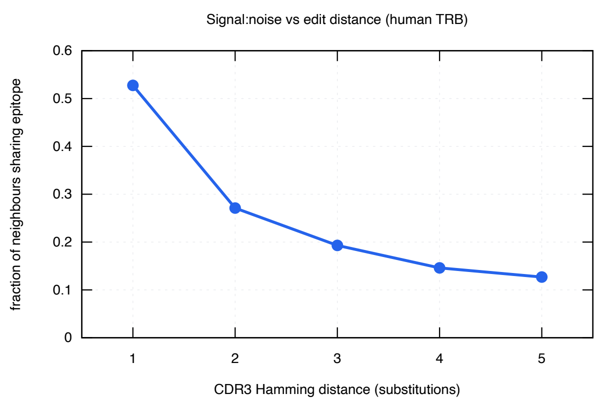

Figure 5: Fraction of CDR3 neighbours sharing an epitope vs Hamming distance (human TRB): distance 1 is the signal:noise optimum [6].

should be down-weighted as noise; instead it is uniform. The weighting is robust rather than cross-V-specific (in relative terms the cross-V gain is larger only because its baseline is far lower).

Germline-loop V-similarity does not recover the prior.. If the same-V advantage were the chemistry of the V-encoded contacts, it should degrade gracefully with germline CDR1+CDR2 similarity, licensing fuzzy V matching (e.g. TRBV5-1≈TRBV5-8). It does not: cross-V co-specificity stays near 12–13% across the whole similarity range, with no monotone gradient and the most-similar bin no higher (Table 3). The V signal is therefore *gene identity*, not contacting-loop similarity — so soft V-clustering by CDR1/CDR2 does not recover it, and a hard or near-hard V constraint remains the right default. (The DE-loop / "CDR2.5" inside FWR3 is not isolated in our germline table; folding it in is a possible refinement, but it is unlikely to bend a flat signal.) The clustering machinery (`match/vgene.py`: germline-loop similarity and single-linkage V groups, 66 → 47 TRB

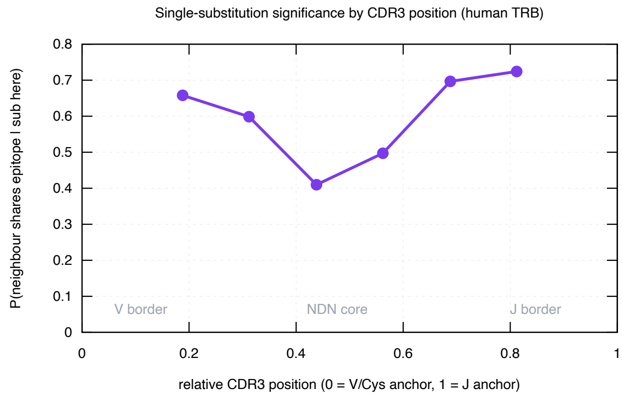

Figure 6: Pr(neighbour shares epitope | single substitution at relative position), human TRB. The NDN core is where a mismatch most changes specificity; the V/J borders are germline noise.

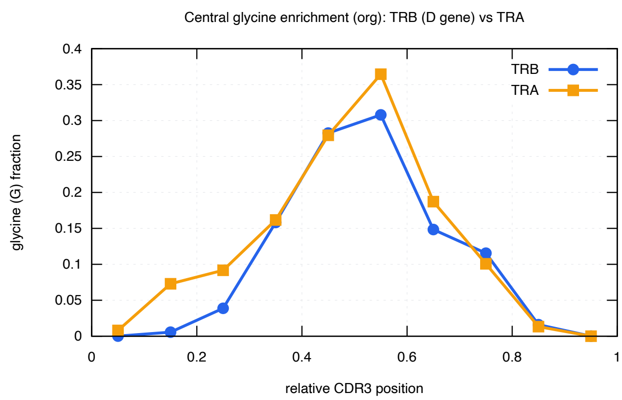

Figure 7: Glycine fraction by relative CDR3 position (human): the NDN core is $\sim$ 30–36% glycine in both TRA and TRB (insertion/D-gene signature).

groups at a 0.8 cut) is shipped for downstream use regardless.

The pattern holds across species, chain and MHC class.. Sweeping every {human,mouse}$\times${TRA,TRB}$\times${M I,MHC-II} cell of VDJdb (`bench/vgene_scan.py`, Fig. 8, Table 4) reproduces both findings everywhere: a V match is a large co-specificity prior (same/cross ratio 1.4–17$\times$), and germline CDR1+CDR2 similarity barely predicts cross-V co-specificity (point-biserial r between -0.06 and $+0.13$ in every adequately powered cell; the single $r = 0.33$ is mouse TRB MHC-II with only 35 cross-V pairs). The near-empty high-similarity bins are themselves telling: germline-similar but non-identical V genes are rare, so CDR1/CDR2 clustering has almost no pairs to act on. We conclude — comprehensively, not just on human TRB — that focusing on CDR1/CDR2 to soften the V match is not supported; V should enter the model as a strong, near-binary prior.

No framework or CDR region recovers it — it is gene identity.. The coarse similarity above leaves two escapes: maybe the signal lives in the framework (FR) rather than CDR1/CDR2, or maybe it is CDR1/CDR2 but — like CDR3 — needs *near-exact* identity that a ratio washes out.

V stratum	pairs	co-specific	flat	+possig	+region
same V	130 273	53.1%	0.671	0.694	0.695
cross V	901 819	12.3%	0.164	0.185	0.182

Table 2: V-gene-stratified leave-one-out retrieval (human TRB, edit distance $\leq$ 4, pooled micro PR-AUC). A V match is a $\sim$4.3$\times$ co-specificity prior; position weighting (+possig, +region) adds $\approx +0.02$ in both strata.

germline CDR1+CDR2 similarity	cross-V pairs	co-specific
$[0.0, 0.3)$	464 817	11.6%
$[0.3, 0.5)$	352 184	13.0%
$[0.5, 0.7)$	66 838	12.8%
$[0.7, 0.9)$	15 831	16.6%
$[0.9, 1.0)$	2 149	8.3%

Table 3: Cross-V co-specificity does not rise with germline contacting-loop similarity (human TRB): the same-V prior (53%) is not interpolated by V-gene similarity, so soft V-clustering does not recover it. The V signal is gene identity, not loop chemistry.

An exact-match decomposition closes both (`bench/vregion_decompose.py`, human TRB MHC-I): among cross-V neighbour pairs we stratify by *which* germline region is identical between the two V genes. No region does it (Table 5): every FR- or CDR-identical stratum sits *at or below* the cross-V floor (22%), nowhere near the same-V level (70%); CDR2 alone is the only one above floor (1.3$\times$) and still far short. Even both CDR1 *and* CDR2 identical does not help (4.5%, $n = 22$), and binning by CDR1+CDR2 edit distance gives a *flat-to-negative* gradient — identical loops 4.5%, distance 1–2 18.7%, 3–5 28.2%, 6+ 22.0% — the opposite of "closer $\Rightarrow$ more co-specific". So there is no decisive FR–CDR combination short of full gene identity: a different V gene with the same CDR1/CDR2 loops still does not behave like the same V. The V prior is the exact gene (its CDR3-germline anchor and integrated docking geometry), not a transferable contacting-loop similarity — which is why V belongs in the model as a near-binary prior, not a soft loop-similarity term. (Robust across scope: subs 1 and subs 2 give the same picture.)

7. Benchmark: the reference-replicated shortlist

The latest VDJdb release is our standing benchmark, but its labels vary in confidence. The cleanest gold standard is a clonotype–epitope association reported independently in ≥ 2 references (`db.replicated`; 1214 TRB and 839 TRA such pairs in the current human release) — the same "shortlist" idea used for `mhcmatch`. We measure annotation accuracy on it by leave-one-out nearest-neighbour voting against the rest of VDJdb at the signal:noise-optimal scope (subs 1), excluding the clonotype's own exact CDR3 so the label must come from *other* TCRs (`bench/shortlist_accuracy.py`). Replicated TCRs are annotated at $\sim$53% top-1 versus $\sim$16% for size-matched single-reference controls of the same epitopes (Table 6) — a $\sim$3$\times$ gap: reference-replicated entries are the public, convergent, learnable TCRs, while single-reference entries are largely private singletons whose epitope is simply not reachable from neighbours (recall $\sim$20–29%). Restricting neighbours to the same V family helps,

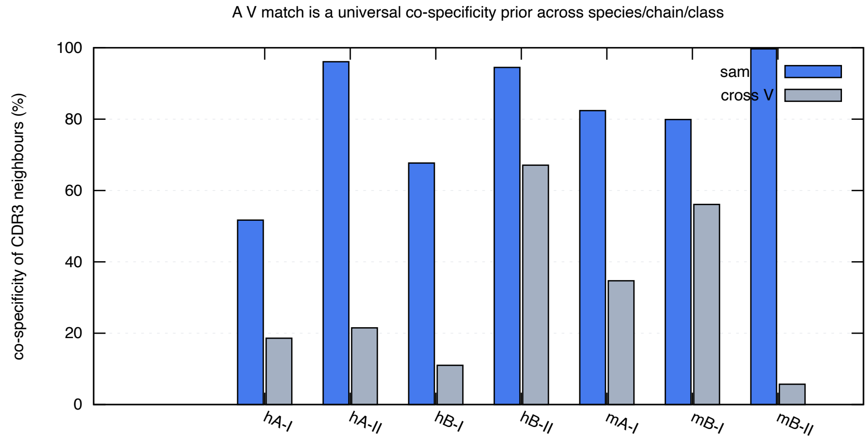

Figure 8: Same-V vs cross-V CDR3-neighbour co-specificity across species/chain/MHC-class cells of VDJdb (h/m = human/mouse; A/B = TRA/TRB; I/II = MHC class). A V match is a universal co-specificity prior; cross-V neighbours are far less likely to share the epitope, and germline CDR1/CDR2 similarity does not bridge the gap (Table 4).

cell (species / chain / class)	same-V	cross-V	ratio	r(vsim, co-spec)
human TRA MHC-I	53.8%	18.4%	2.9	0.086
human TRA MHC-II	96.1%	10.2%	9.4	0.126
human TRB MHC-I	69.0%	9.7%	7.1	0.046
human TRB MHC-II	94.2%	67.2%	1.4	0.039
mouse TRA MHC-I	82.8%	34.8%	2.4	0.047
mouse TRB MHC-I	80.6%	56.3%	1.4	−0.056
mouse TRB MHC-II	99.7%	5.7%	17.4	0.331

Table 4: Per-cell same-V vs cross-V co-specificity and the point-biserial correlation between germline CDR1+CDR2 similarity and co-specificity among cross-V pairs (unit-cost candidates, top 6 epitopes/cell, $\geq$ 30 CDR3/epitope; mouse TRA MHC-II omitted, n too small). The V prior holds everywhere; $r \approx 0$ wherever it is well powered.

and helps *more* at the wider subs 2 scope (54.1 → 58.4% for TRB) — exactly where cross-V false neighbours appear (§ 6), confirming the V prior is worth its place in the model.

8. Worked examples

Pairwise (different lengths).. `seqtree` aligns CDR3s of differing length, placing indels where the loop genuinely lengthens. The edit-distance alignments below (influenza GIL-specific TRB CDR3s) show the conserved `CASS` V-anchor and `DTQYF` J-motif with substitutions and indels confined to the central NDN core:

```
query (ref): CASSPRSGETQYF   epitope GILGFVFTL   (unit / edit-distance alignment)


vs CASSNRSGETQYF  (len 13; subs=1 ins=0 dels=0; CIGAR 4=1X8=)
```

cross-V stratum (human TRB MHC-I)	co-specific	n
cross-V baseline	22.4%	40 388
FR1 identical	14.3%	328
CDR1 identical	16.3%	760
FR2 identical	13.3%	300
CDR2 identical	28.6%	584
FR3 identical	15.5%	200
CDR1 & CDR2 identical	4.5%	22
all FR identical	15.5%	200
same-V reference	69.5%	16 492

Table 5: Cross-V co-specificity stratified by which germline region is identical between the two V genes (edit distance ≤ 2, exact self excluded). No framework or CDR region — nor CDR1 $\wedge$ CDR2 together — lifts co-specificity toward the same-V level; the decisive feature is gene identity itself.

chain	set	n	top-1 (CDR3)	top-1 (V-restr.)	recall
TRA	shortlist ($\geq$ 2 ref)	839	52.1%	53.3%	62.3%
TRA	control (1 ref)	839	15.9%	15.0%	29.1%
TRB	shortlist ($\geq$ 2 ref)	1214	53.3%	54.4%	57.7%
TRB	control (1 ref)	1214	17.2%	15.7%	19.7%

Table 6: Leave-one-out epitope-annotation accuracy on the reference-replicated shortlist vs a size-matched single-reference control (human, subs 1, exact self excluded). Top-1 = nearest-neighbour-vote epitope correct; recall = true epitope reachable within scope.

```
  CASSPRSGETQYF
  |||| ||||||||
  CASSNRSGETQYF

vs CASSIRSGESTQYF  (len 14; subs=1 ins=1 dels=0; CIGAR 4=1X4=1I4=)
  CASSPRSGE-TQYF
  |||| |||| ||||
  CASSIRSGESTQYF

vs CASSLFSETQYF  (len 12; subs=2 ins=0 dels=1; CIGAR 4=2X1=1D5=)
  CASSPRSGETQYF
  ||||  | |||||
  CASSLFS-ETQYF

vs CASSLGETQYF  (len 11; subs=1 ins=0 dels=2; CIGAR 4=2D1X6=)
  CASSPRSGETQYF
  ||||   ||||||
  CASS--LGETQYF

vs CASSPWSGSQETQYF  (len 15; subs=1 ins=2 dels=0; CIGAR 5=1X2=2I5=)
  CASSPRSG--ETQYF
  ||||| ||  |||||
```

```
CASSPWSGSQETQYF
```

Multiple (same length).. Aligning same-length same-epitope CDR3s (no indels needed) makes the conservation profile explicit (Figs. 9–10): columns are shaded by conservation, dark at the germline-determined `CASS` prefix and J motif, light through the variable NDN core. This is the structural counterpart of the finding of Shcherbinin et al. [5] that specificity-preserving CDR3 substitutions occur predominantly outside the peptide/MHC-contacting positions while preserving the loop backbone — i.e. the tolerated variation we score in the NDN core is exactly the variation that does not break recognition.

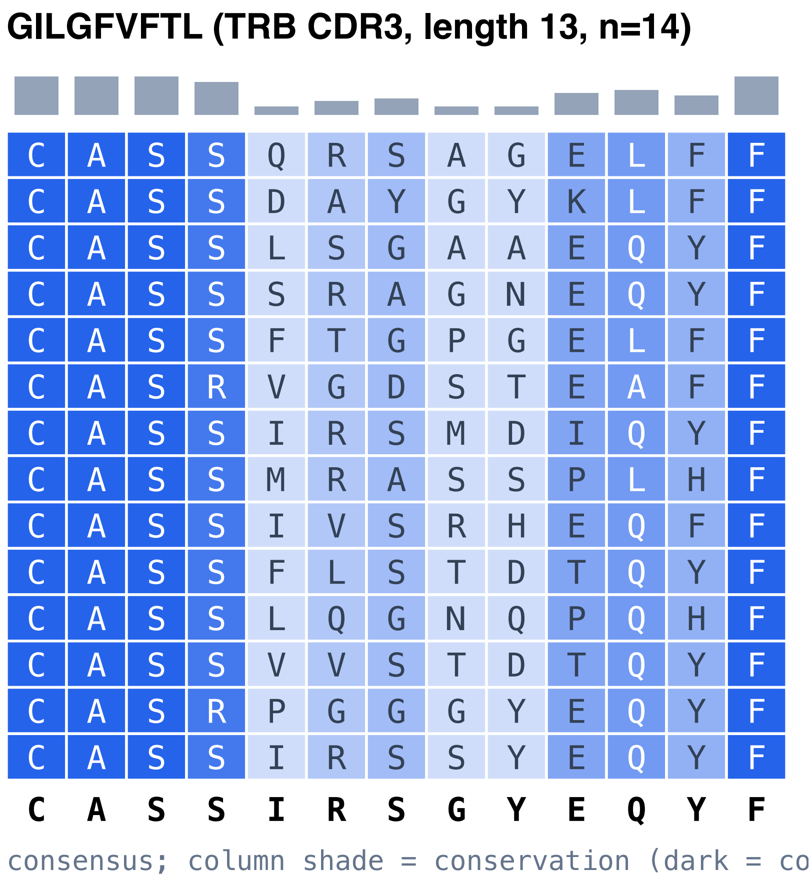

Figure 9: GIL (influenza M1, A*02) TRB CDR3 block; conserved V/J flanks, variable NDN core.

9. Limitations and roadmap

The data-derived NDN matrix is estimated from a limited set of same-antigen single-substitution pairs and does not beat BLOSUM62 at any scope; region-aware germline+trimming weighting gives a small, consistent gain over the flat matrix but not over BLOSUM62. The substitution *alphabet* is thus a second-order lever. What does move the needle is *position*: reweighting BLOSUM62 by the

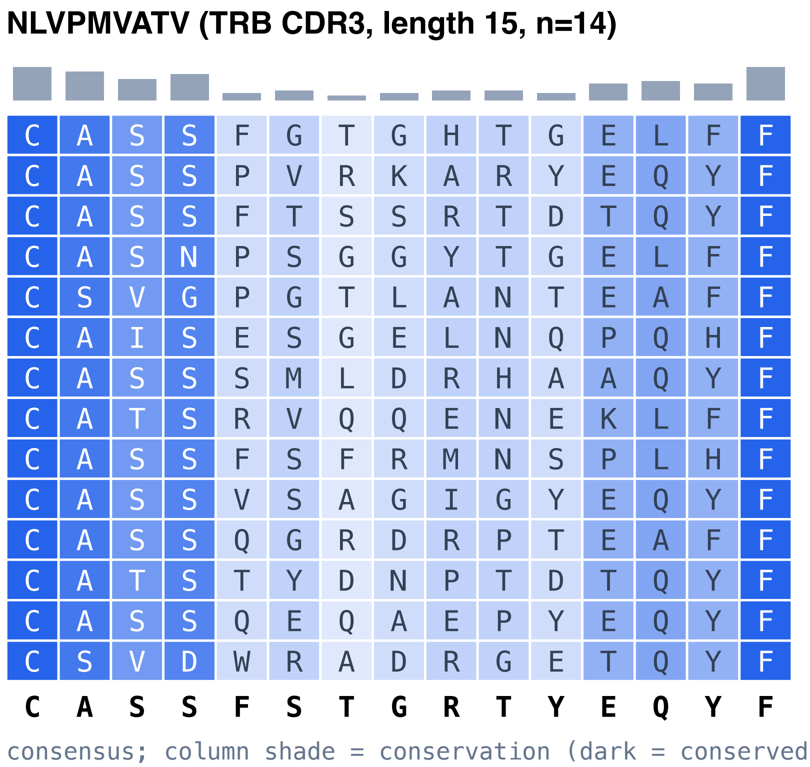

Figure 10: NLV (CMV pp65, A*02) TRB CDR3 block; same conservation pattern.

central-significance PSSM beats flat BLOSUM62 at every held-out epitope (§ 5), confirming that for CDR3 *where* a mismatch falls matters more than *which* residue it is. The *V-gene* dimension (§ 6) turns out to be a strong but near-binary lever: a V match is a $\sim$4.3$\times$ co-specificity prior, yet it does *not* interpolate with germline contacting-loop similarity, so soft V-clustering by CDR1/CDR2 does not recover it and a near-hard V constraint stays the default. The open work is therefore to model V as a strong prior in a joint V+CDR3 E-value rather than to soften the V match, plus a per-chain treatment (TRA vs TRB). Separately, the control's V–J usage must match the query repertoire's or the null is mis-specified — whether to renormalize V–J usage (at the risk of erasing genuine V-gene bias) is an open question to settle on held-out epitopes.

A further direction is to predict, *a priori*, which epitopes are intrinsically easy versus hard to annotate — the wide spread of per-epitope PR-AUC in our experiments (e.g. the convergent, "featured" CMV NLV vs the diffuse, "featureless" influenza GIL) is not noise but biology. A pMHC with prominent TCR-facing features selects a focused, motif-rich repertoire that clusters cleanly, whereas a featureless one is recognised by structurally diverse TCRs in varied docking modes and yields no tight motif [8, 9, 7, 2]; clustering-based specificity inference succeeds or fails accordingly [1]. A model that scores epitope "annotability" from repertoire convergence (and, where available, pMHC structural features) would let us report calibrated confidence per epitope and triage GIL-like hard cases — a planned addition to the benchmark. These extensions, and the full comparison against existing tools, are ongoing work.

References

[1] Dan Hudson, Alex Lubbock, Mark Basham, and Hashem Koohy. A comparison of clustering models for inference of T cell receptor antigen specificity. *Immunoinformatics*, 13:100033, 2024.

[2] Hung-Pin Peng, Kuo Hao Lee, Jhih-Wei Jian, and An-Suei Yang. Origins of specificity and affinity in antibody-protein interactions. *Proceedings of the National Academy of Sciences of the United States of America*, 111(26):E2656–E2665, 2014.

[3] Anna Postovskaya, Koen Vercauteren, Pieter Meysman, and Kris Laukens. tcrBLOSUM: an amino acid substitution matrix for sensitive alignment of distant epitope-specific TCRs. *Briefings in Bioinformatics*, 26(1):bbae602, 2024.

[4] Dmitrii S. Shcherbinin, Vlad A. Belousov, and Mikhail Shugay. Comprehensive analysis of structural and sequencing data reveals almost unconstrained chain pairing in TCR$\alpha\beta$ complex. *PLoS Computational Biology*, 16(3):e1007714, 2020.

[5] Dmitrii S. Shcherbinin, Vadim K. Karnaukhov, Ivan V. Zvyagin, Dmitriy M. Chudakov, and Mikhail Shugay. Large-scale template-based structural modeling of T-cell receptors with known antigen specificity reveals complementarity features. *Frontiers in Immunology*, 14:1224969, 2023.

[6] Mikhail Shugay, Dmitriy V. Bagaev, Ivan V. Zvyagin, Renske M. Vroomans, Jeremy Chase Crawford, Garry Dolton, et al. VDJdb: a curated database of T-cell receptor sequences with known antigen specificity. *Nucleic Acids Research*, 46(D1):D419–D427, 2018.

[7] InYoung Song, Anna Gil, Rabinarayan Mishra, Dario Ghersi, Liisa K. Selin, and Lawrence J. Stern. Broad TCR repertoire and diverse structural solutions for recognition of an immunodominant CD8+ T cell epitope. *Nature Structural & Molecular Biology*, 24(4):395–406, 2017.

[8] Stephen J. Turner, Katherine Kedzierska, Helen Komodromou, Nicole L. La Gruta, Michelle A. Dunstone, Andrew I. Webb, Richard Webby, Helen Walden, Weidong Xie, James McCluskey, Anthony W. Purcell, Jamie Rossjohn, and Peter C. Doherty. Lack of prominent peptide-major histocompatibility complex features limits repertoire diversity in virus-specific CD8+ T cell populations. *Nature Immunology*, 6(4):382–389, 2005.

[9] Xinbo Yang, Guobing Chen, Nan-Ping Weng, and Roy A. Mariuzza. Structural basis for clonal diversity of the human T-cell response to a dominant influenza virus epitope. *The Journal of Biological Chemistry*, 292(45):18618–18627, 2017.